

Data Science Techniques and Real-World Applications

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WS 2026 Agenda

Course Introduction

Data Science Techniques and Real-World Applications is a practical, hands-on course centered on how to approach data and analyze it in a responsible and sound way. Students build core data science skills such as data wrangling, statistical reasoning, causal thinking, and predictive modeling, and apply them to real business case studies, while gaining practical, non-hype exposure to Generative AI, Agentic AI, and cloud-based data science applications. The course combines instructor-led sessions, guest speakers, and student group presentations on current topics in applied data science.

Day 1 – Thursday 3 September 2026

Block	Time	Content
1	09:30–11:00	Introduction Team formation Presentation-topic overview Python fundamentals
2	11:15–12:45	Python fundamentals: Classes, Functions, NumPy

Day 2 – Thursday 10 September 2026

Block	Time	Content
1	09:30–11:00	Python fundamentals: Pandas: tidy data, merges, and fuzzy/regex entity matching (feeds CS1)
2	11:15–12:45	Data visualization (Matplotlib/Seaborn) + Exploratory Data Analysis
3	14:00–15:30	Statistics & hypothesis testing — what a number can and can't tell you
4	15:45–17:15	Applied practice lab → CS1 assigned (due Sun 27 Sept)

Day 3 — Thursday 17 September 2026

Block	Time	Content
1	09:30–11:00	Regression analysis — fitting and interpreting coefficients (simple/multiple regression)
2	11:15–12:45	Causality — correlation vs. causation, confounding: why the coefficient you just interpreted needs more scrutiny before you act on it
3	14:00–15:30	Guest speaker #1
4	15:45–17:15	Presentations — Block A

Gap week — no class Thursday 24 September 2026

Day 4 — Thursday 1 October 2026

Block	Time	Content
1	09:30–11:00	ML foundations I — gradient descent, logistic regression
2	11:15–12:45	ML foundations II — decision trees, evaluation metrics, calibration
3	14:00–15:30	Generative AI I — how LLMs actually work, capabilities & limits
4	15:45–17:15	Generative AI II — prompting, RAG, responsible use (hallucination, bias, cost) → CS2 assigned (due Sun 18 Oct)

CS2 assigned today (bank-marketing / H2O / calibration case), due 18 Oct.

Day 5 — Thursday 8 October 2026

Block	Time	Content
1	09:30–11:00	Agentic AI I — what agents are, tool use, the agent loop, permissions & verification
2	11:15–12:45	Agentic AI II — hands-on: using a coding/research agent on a small data task
3	14:00–15:30	Presentations — Block B
4	15:45–17:15	Presentations — Block C → CS3 assigned (due Sun 25 Oct)

CS3 assigned today (build-your-own-agent case, new), due 25 Oct (after course ends).

Day 6 — Thursday 15 October 2026

Block	Time	Content
1	09:30–11:00	Cloud/AWS — fundamentals
2	11:15–12:45	Cloud/AWS — fundamentals
3	14:00–15:30	Presentations — Block D
4	15:45–17:15	Presentations — Block E + course wrap-up

Case studies (3 total, all ≥ 2 weeks)

Case	Topic	Assigned	Due	Gap
CS1	Fuzzy-merge entity resolution (SE/CRSP), reused from WS2025	Thu 10 Sept	Sun 27 Sept	17 days
CS2	Bank marketing: classification, probability calibration, H2O	Thu 1 Oct	Sun 18 Oct	17 days
CS3	Build-your-own-agent (new for WS2026)	Thu 8 Oct	Sun 25 Oct	17 days

Presentation topics (Only suggestions — students may also propose their own)

1. Web scraping & data acquisition
2. From classic NLP to transformers (tokenization/TF-IDF $\rightarrow$ modern models)
3. Recommendation systems
4. Hugging Face & the open-model ecosystem
5. AI agents in the wild (coding assistants, customer service, research agents)
6. Cloud & MLOps: deploying data science in production
7. Responsible & ethical AI (bias, privacy, EU AI Act)
8. AI in a specific industry vertical (finance/healthcare/marketing)
9. Model monitoring & evaluation in production

Course Evaluation

- Grading is based on case studies and the group presentation.
- Weighting: Case studies (3) — 75% — 90 points | Group presentation — 25% — 30 points.
- Group presentations will be graded by all of us; therefore:
 - **Attendance is mandatory for all student presentation blocks and both guest-speaker sessions.** Unexcused absence results in a point deduction from the final grade.